Supplementary Material

Impact of Air-Cathodes on Operational Stability of Single-Chamber Microbial Fuel Cell Biosensors for Wastewater Monitoring

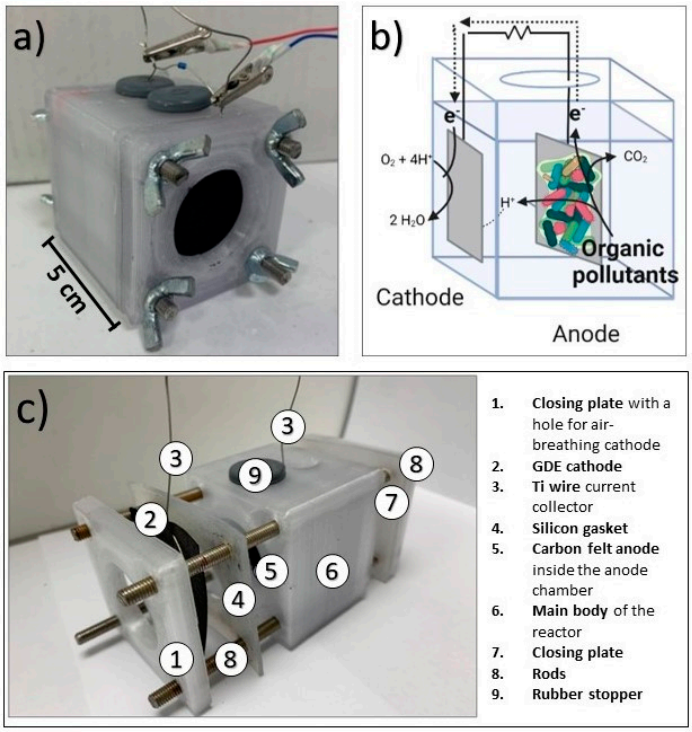


Figure S1 a) MFC reactor utilized during the study; (b) schematic representation of the structure of a SC MFC with an air cathode, and (c) open reactor displaying the different components.

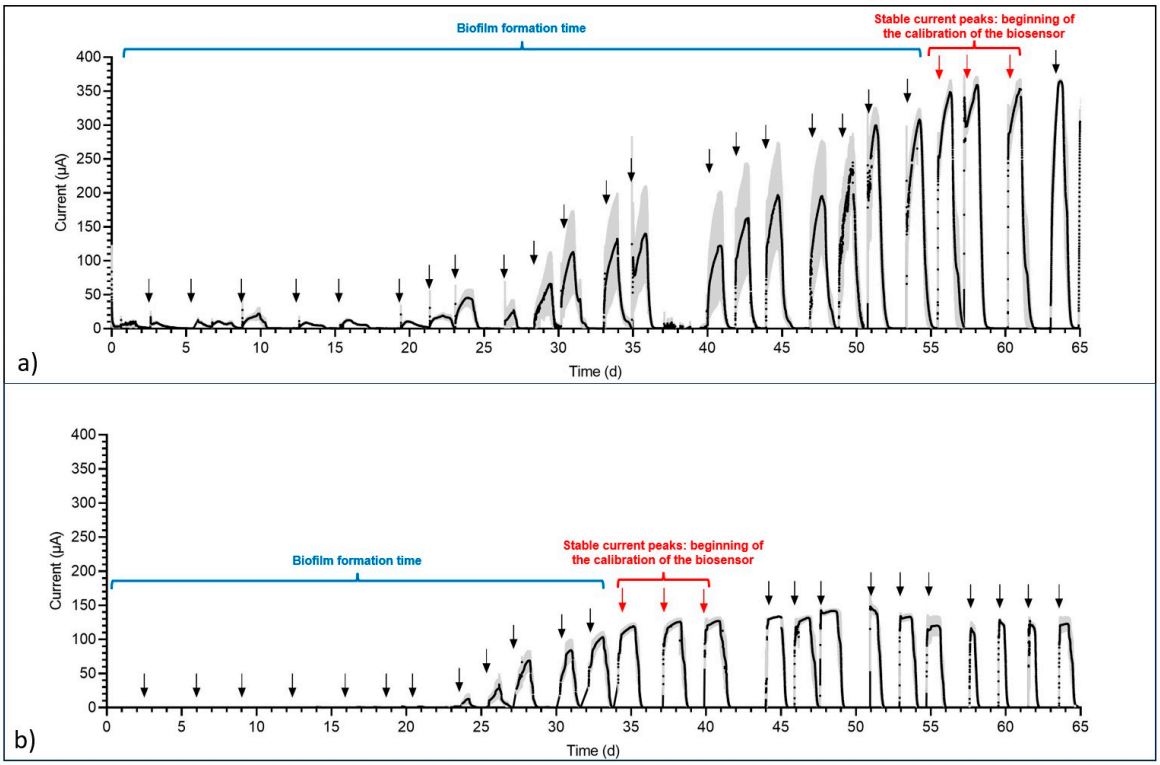


Figure S2 Current produced during the biofilm formation time in a) CC-MFC and b) CP-MFC with external resistor of 1000 Ω. The arrows indicate the anolyte replacement, which contains 0.6 mg L-1 Na-acetate. Data are shown as mean of 4 replicates ± Standard Error of the Mean.

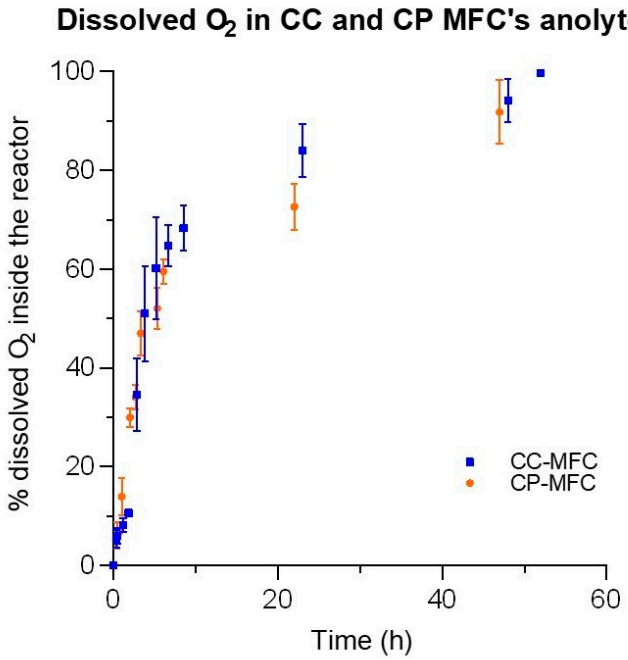


Figure S3 Concentration of dissolved oxygen in the electrolyte of MFC reactors equipped with CC- and CP cathode over time. Time 0 h corresponds to replacing of the anolyte inside the aerobic chamber. Over time, O2 diffused inside the reactor, and the concentration of dissolved O2 in the anolyte increased and it was measured with a Fiber-Optic Oxygen Meter and a Pyro Oxygen Logger Software (FireSting O2, Aachen, Germany). Data are presented as mean of 2 replicates ± SD.

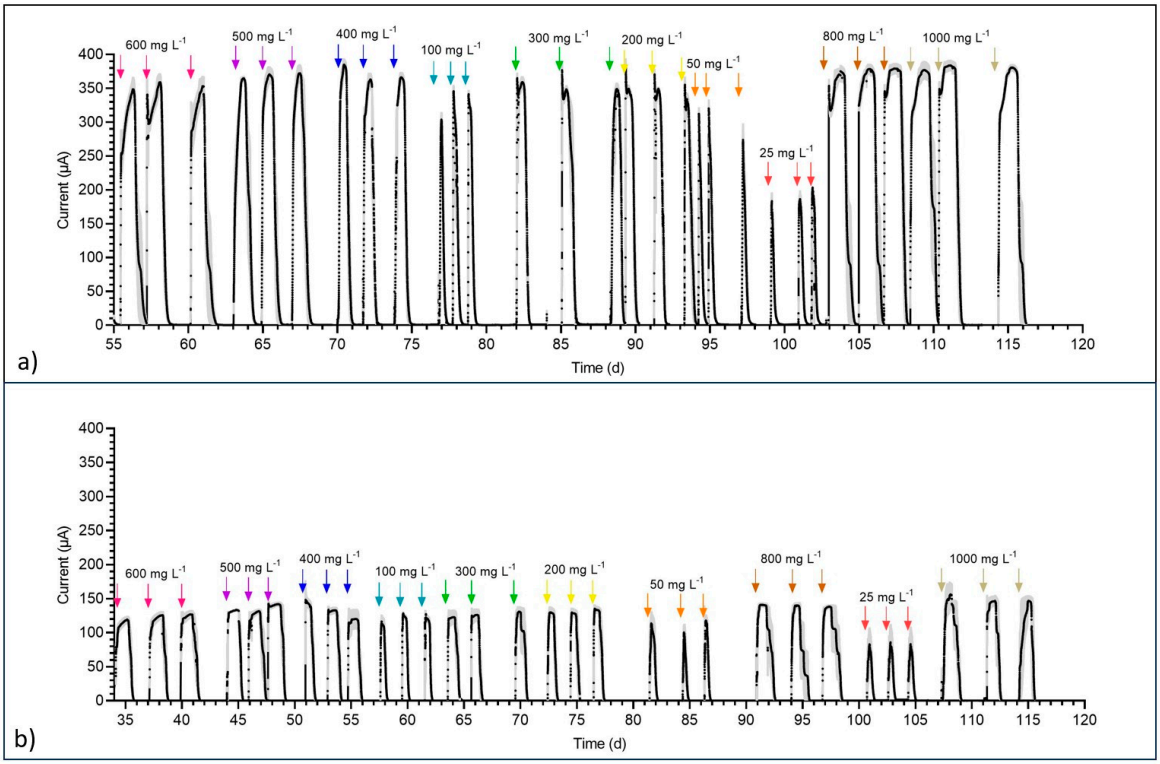


Figure S4 Current produced during the calibration of the biosensor with different Na-acetate concentrations in a) CC-MFC and b) CP-MFC with external resistor of 1000 Ω. The anolyte replacement is marked by arrows, with each arrow's color representing a different Na-acetate concentration in the media, as annotated in the graph. Each Na-acetate concentration underwent three tests to evaluate the reproducibility of the biosensor. Data are shown as mean of 4 replicates ± Standard Error of the Mean.

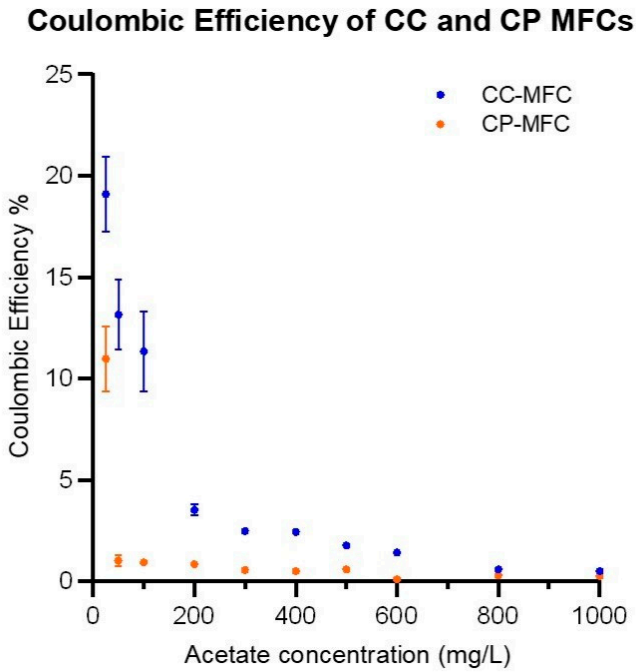


Figure S5 Correlation between the coulombic efficiency produced by the CC-MFC (blue) and CP-MFC (orange) and the acetate concentration of the feeding solution. Data are shown as mean of 4 replicates ± Standard Error of the Mean.

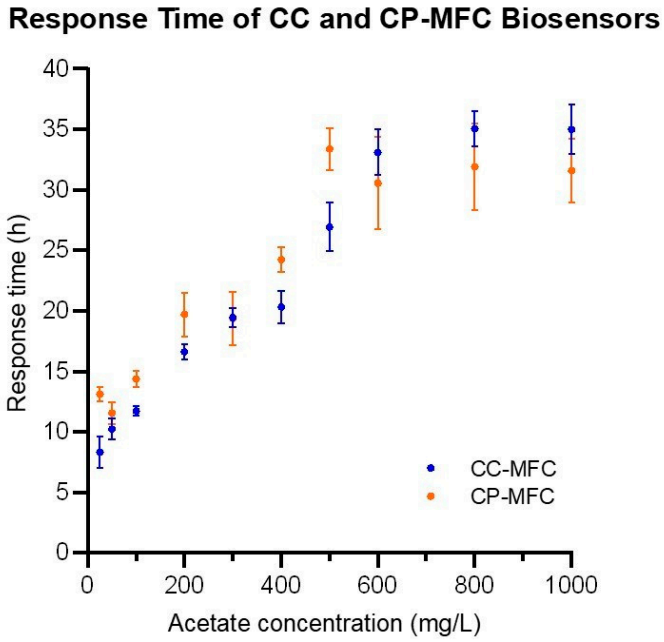


Figure S6 Correlation between the response time of CC-MFC (blue) and CP-MFC (orange) biosensors and the acetate concentration of the feeding solution. Data are shown as mean of 4 replicates ± Standard Error of the Mean.

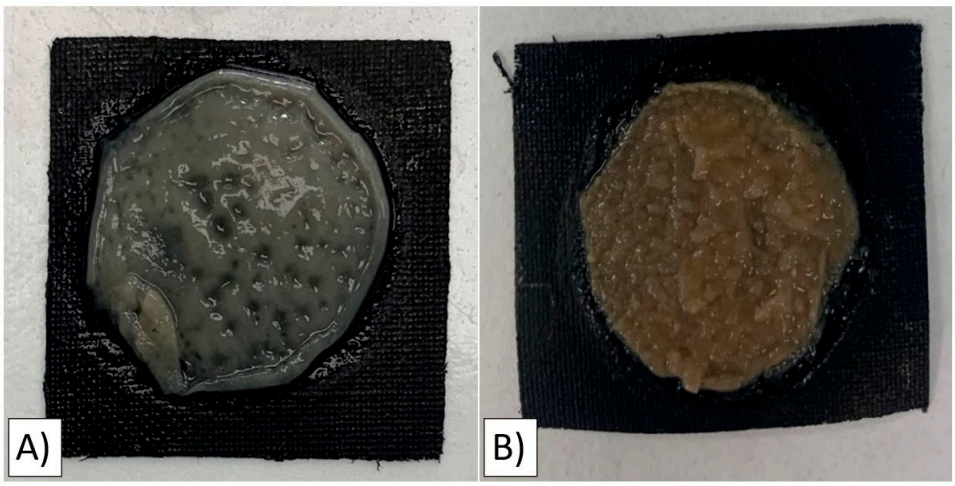


Figure S7 Inner surface of (a) CP GDE cathode at the end of the experiment and of (b) CC GDE cathode at the end of the experiment.

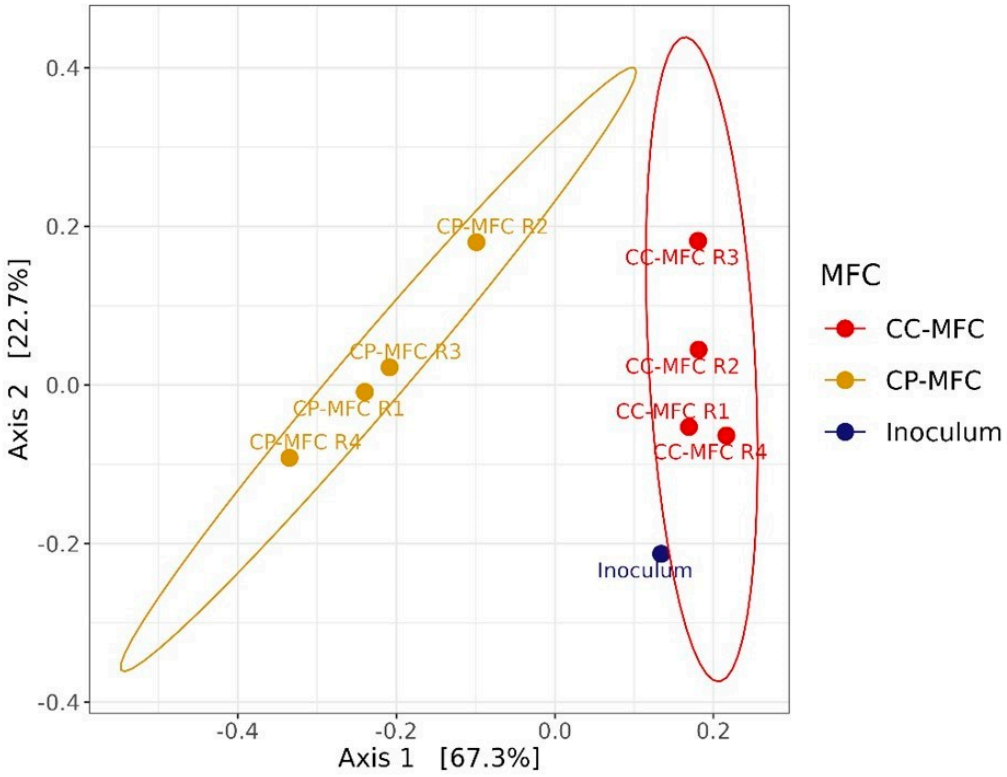


Figure S8 Principal Coordinate Analysis of four replicates of CC-MFCs, CP-MFCs, and the initial inoculum. The analysis, performed with the R package "phyloseq," is supplemented with weighted UniFrac distance assessment. The taxa were filtered above 0.025% .

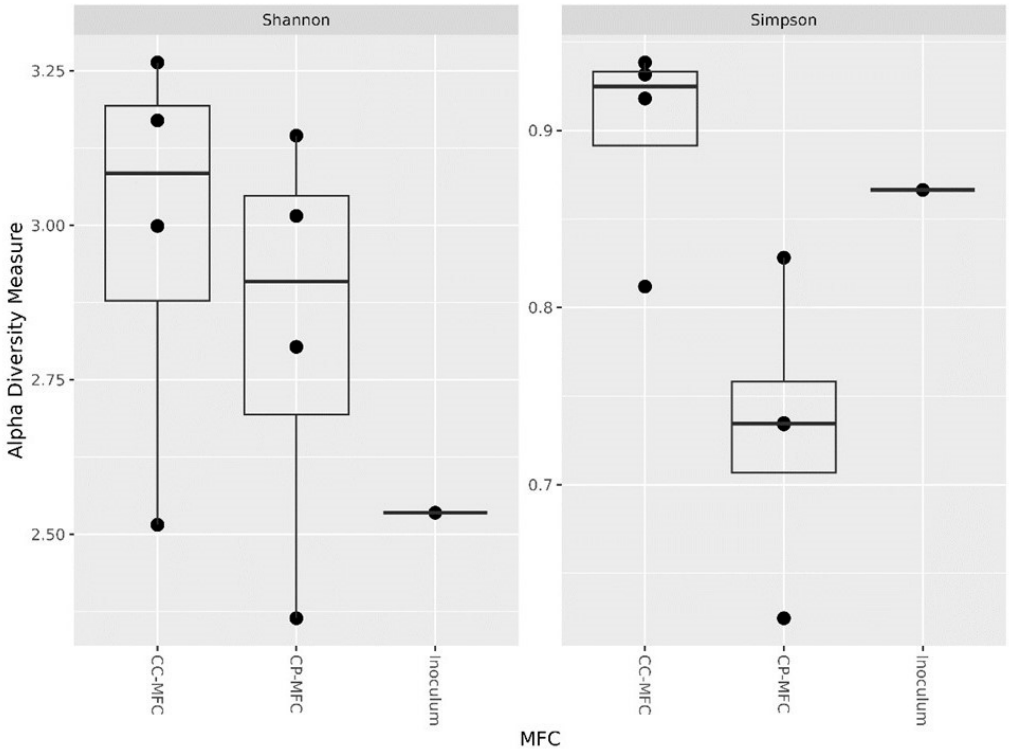


Figure S9 Alpha diversity analysis results using Shannon and Simpson indices for four replicates of CC-MFCs, CP-MFCs, and the initial inoculum. The analysis, performed with the R package "phyloseq," is supplemented with weighted UniFrac distance assessment.

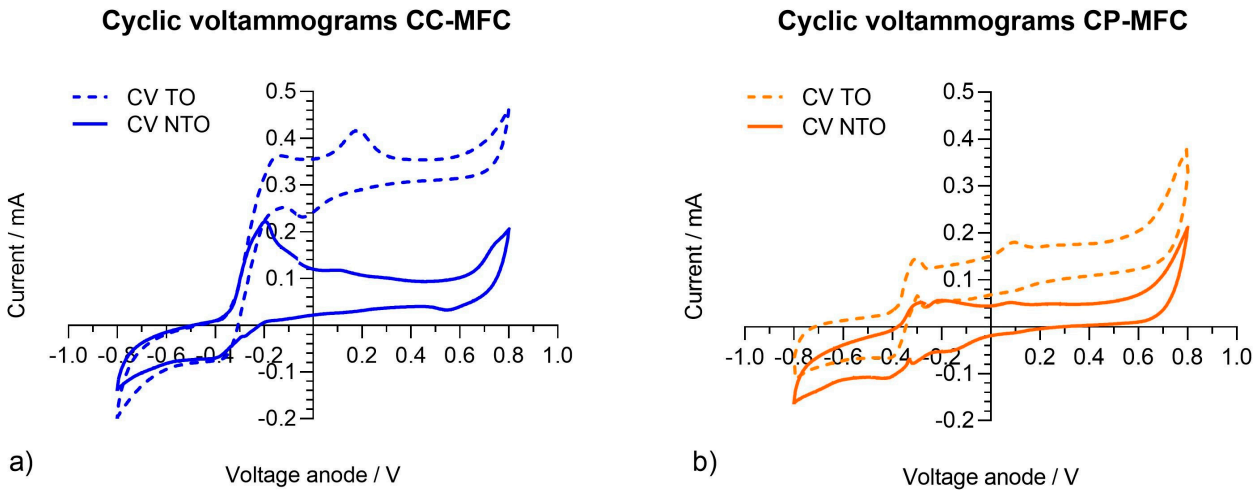


Figure S10 Cyclic voltammograms of (a) CC-MFC and (b) CP-MFC at the end of the experiment with 0.6 g L-1 Na-acetate in minimal medium (TO conditions, dotted line) and acetate-free minimal medium (NTO conditions, continuous line).